

Senior/Lead Linux Kernel Engineer**Sample Job Description**

We are looking for a virtualisation/senior Linux kernel engineer, with at least four years of experience:

- Ability to port the Linux kernel to a new CPU architecture
- Expert in virtualisation technologies such as Xen, L4 or UML is a plus
- Ability to design, architect, develop on Linux kernel internals such as file system code, memory management, scheduler.
- Expert C programmer
- Experience in one CPU architecture and its virtual memory concepts
- Experience in one of the assembly languages such as Intel, ARM, MIPS, PowerPC or similar.
- Ability to use open source software development tools such as git, gcc cross-compiler toolchains
- Experience in embedded systems.
- Experience in SMP and concurrency.
- Fluency in English

Responsibilities:

- This is an exciting hands-on work experience, since you are going to be developing on the core subsystems of an OS kernel and virtualisation software. Since the software is open, your name will take its place as one of the core developers.
- You will be asked to help in porting Linux to a new architecture, write low-level file system code, work on enhancing memory management software, the scheduler, paging, contribute in modifying kernel configuration and build system documentation and testing.

Major employers in India

EMC (VM ware)
 Microsoft
 Cisco Systems (India)
 NetApp
 ObjectWin Technology, Inc
 Yahoo India

Mid career switch

The flexibility to switch from this field lowers as you spend more time in the industry. "It's best to switch at the beginning, as after that one becomes more specialised and more suited to this industry's needs than anywhere else. Though it is possible to switch within the server, storage and network virtualisation industries." Divya says. However, what happens in case one wants to change one's career mid-way after being a part of virtualisation for a good 5-6 years? Makkar says a switch within the industry is possible, but beyond this it can be a little difficult.

Resources for further reading

- <http://www.virtualization.info/predictions/>
- <http://www.expresscomputeronline.com/20040524/technology01.shtml>
- <http://www.info4security.com/story.asp?storyCode=4122239§ioncode=50>
- http://www.cio-weblog.com/50226711/nasscom_on_chinas_it_software_services_industry_evolution.php
- http://www.nasscom.org/download/Annual_Report.pdf
- <http://www.nasscom.in/upload/52333/Knowledge%20Services%20from%20Rural%20Centers.pdf>
- <http://www.nasscom.in/Nasscom/templates/NormalPage.aspx?id=5000>